

From Oracle Legacy to AI-Ready: A Blueprint

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From Oracle Legacy to AI-Ready: A Blueprint

What this session is about and what it isn't...



Database Modernization

- How to safely move data to cloud-native platforms
- How to efficiently design cloud-native databases



AI Implementation

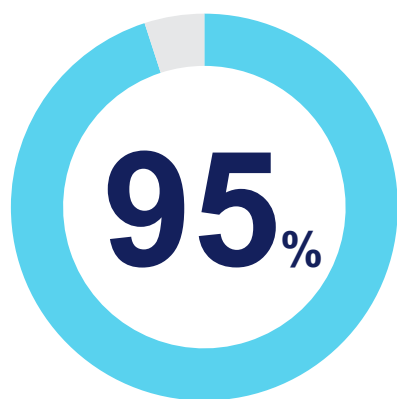
- Creating RAGs
- Vector Databases



AI-Ready Data

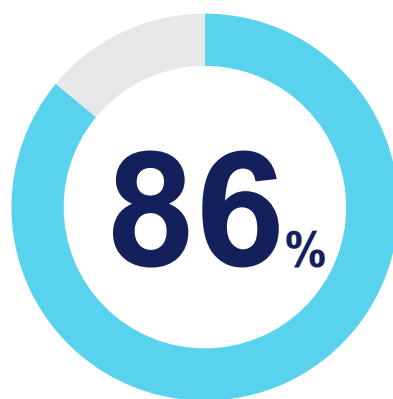
- How to get to a place where you are ready to use your Oracle data in AI

Greatest impediments to AI success come from data management & identity security



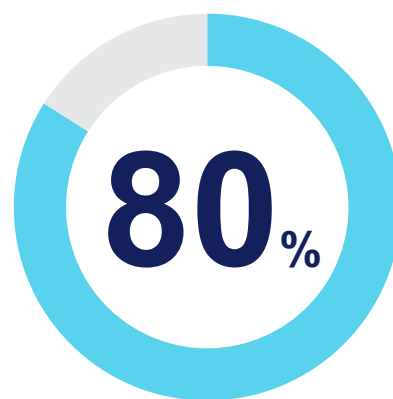
AI and ML projects encounter **data quality issues**

- MIT



Companies experienced **AI related security incidents in last year**

- Cisco's 2025 Cybersecurity Readiness Index



Companies faced **increased security risks from unmanaged devices**

- Microsoft Digital Defense Report

~**180**_{ZB}

Growth of sensitive data and users in 2025 (3x 2020 levels)

Where you are now can dictate where you go next

A benefit of legacy Oracle data is its structure



Schema-on-write

- Pre-defined schema structure
- Familiar approach
- Data is already structured



Schema-on-read

- Schema is defined when reading the data
- More flexible
- Suitable for unstructured data



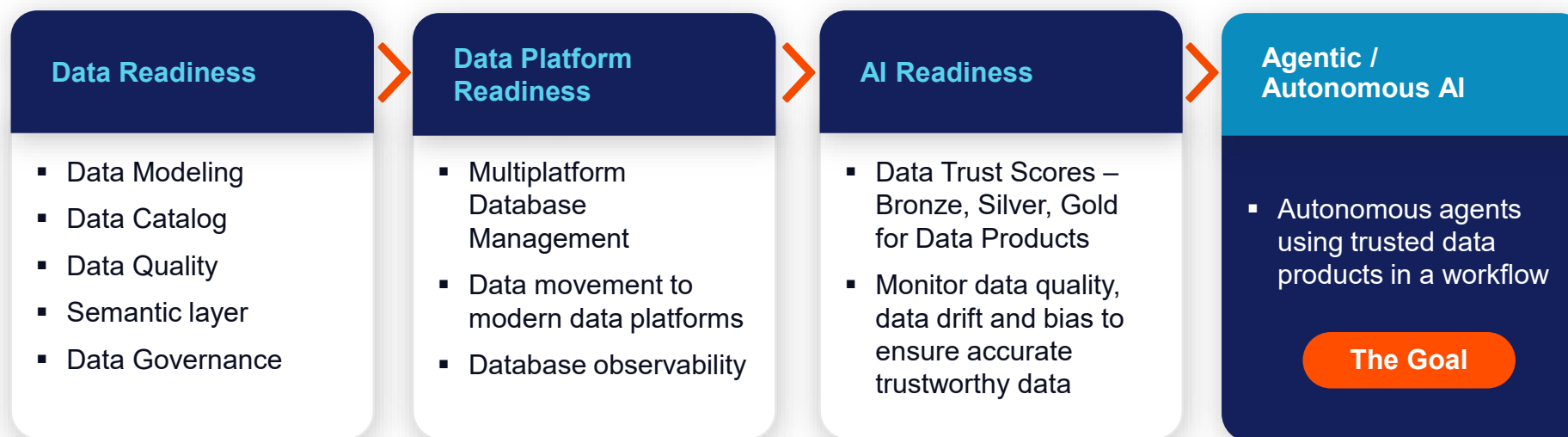
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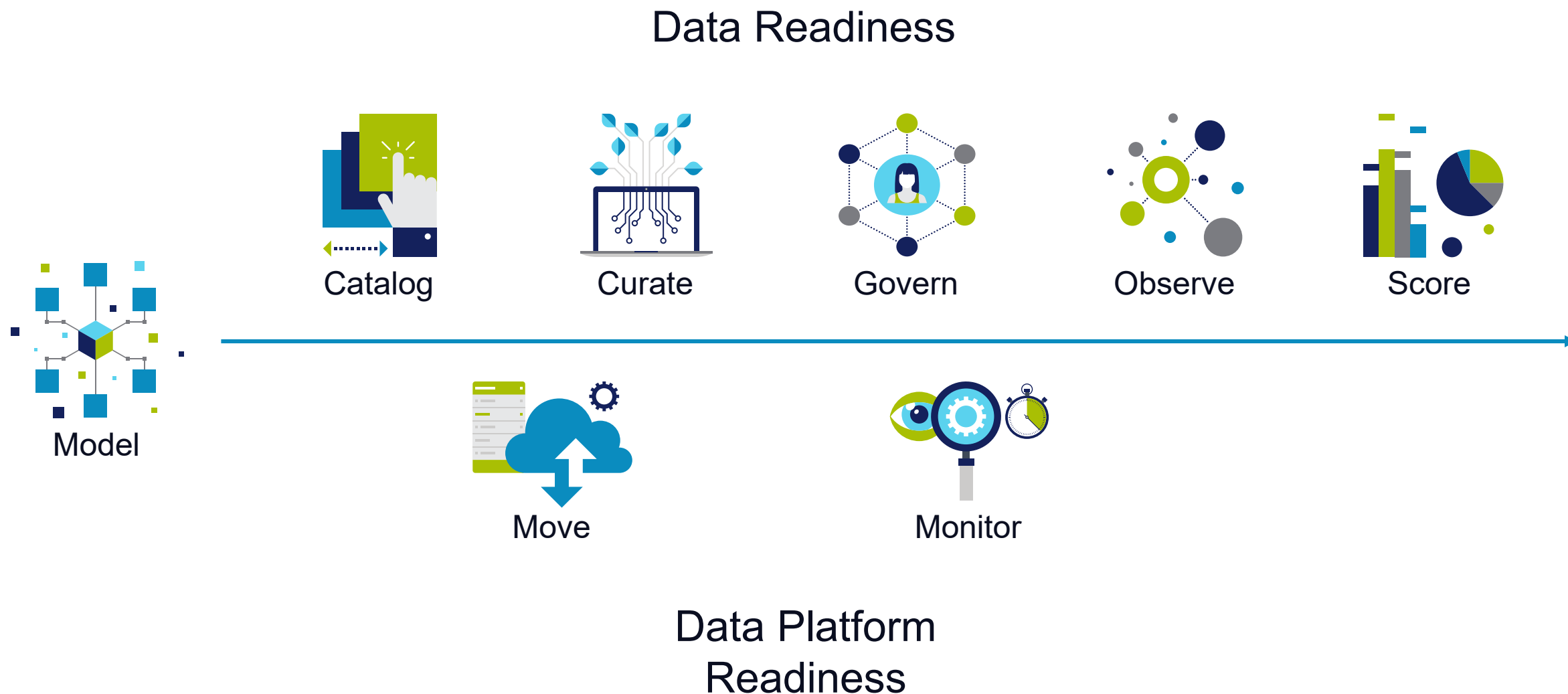
The Blueprint

”

Trusted data solutions for AI success

Building trusted, high-quality data products through data modeling, intelligence, and governance





Data Modeling



**Migrating Data?
The Real Challenge
Isn't Movement,
It's Understanding.**



Reduce Costs Improve Performance Accelerate Time-to-Value

Model Data: Shift Left



without proper data modeling, cloud-native databases turn into expensive, hard-to-govern data swamps with unpredictable costs.

"Honestly, our data warehouse is a mess. We spent a fortune on it, but it's so slow and complex that nobody wants to use it. We're basically back to using spreadsheets." - **r/dataengineering**

"We implemented a Data Lakehouse with all the bells and whistles, but it's become a data swamp. Data quality is all over the place, and finding anything useful is like looking for a needle in a haystack." - **r/bigdata**

"[Platform Y] is great for querying, but the costs can quickly spiral out of control if you're not careful. We're constantly trying to optimize our queries and storage to keep the bills down." - **r/snowflake**

"We're struggling with data governance in [Platform Y]. It's hard to keep track of all the different users and roles, and we're worried about data breaches." - **r/dataengineering**

"[Platform X] is powerful, but it's also complex. We're finding it difficult to manage and maintain, and our team is spending a lot of time troubleshooting issues." - **r/databricks**

"We're not getting the performance we expected from [Platform X]. Our Spark jobs are slow, and we're not sure how to optimize them." - **r/dataengineering**

"[Platform X] promised us a unified data platform, but it's been a nightmare. Integration is a pain, and we're constantly running into compatibility issues." - **r/dataengineering**

"We're locked into [Platform Y], and it's holding us back. It's expensive, inflexible, and doesn't support the latest technologies." - **r/cloudcomputing**



Model

For AI to transform your business, you need a strong **Foundation for AI**



Reverse Engineer

- Create Conceptual, Logical & Physical ER Diagrams of existing Oracle schemas
- Create custom datatype mappings for cloud-native database



Derive & Optimize

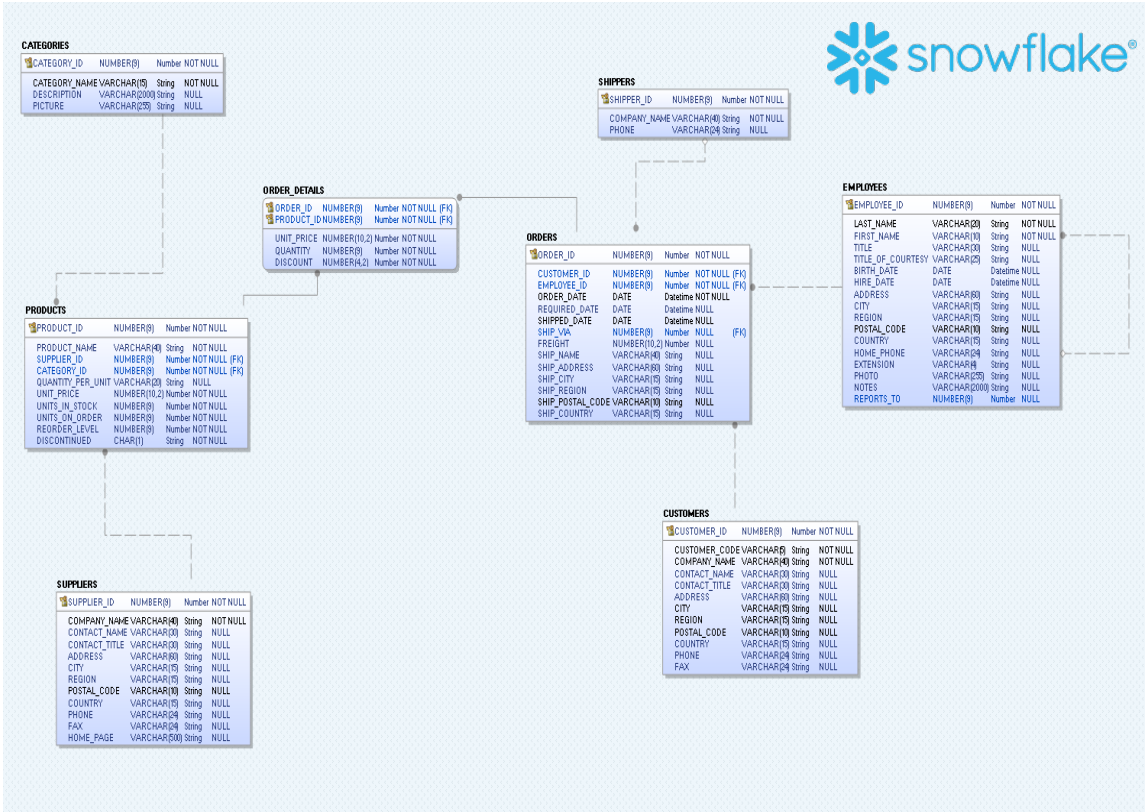
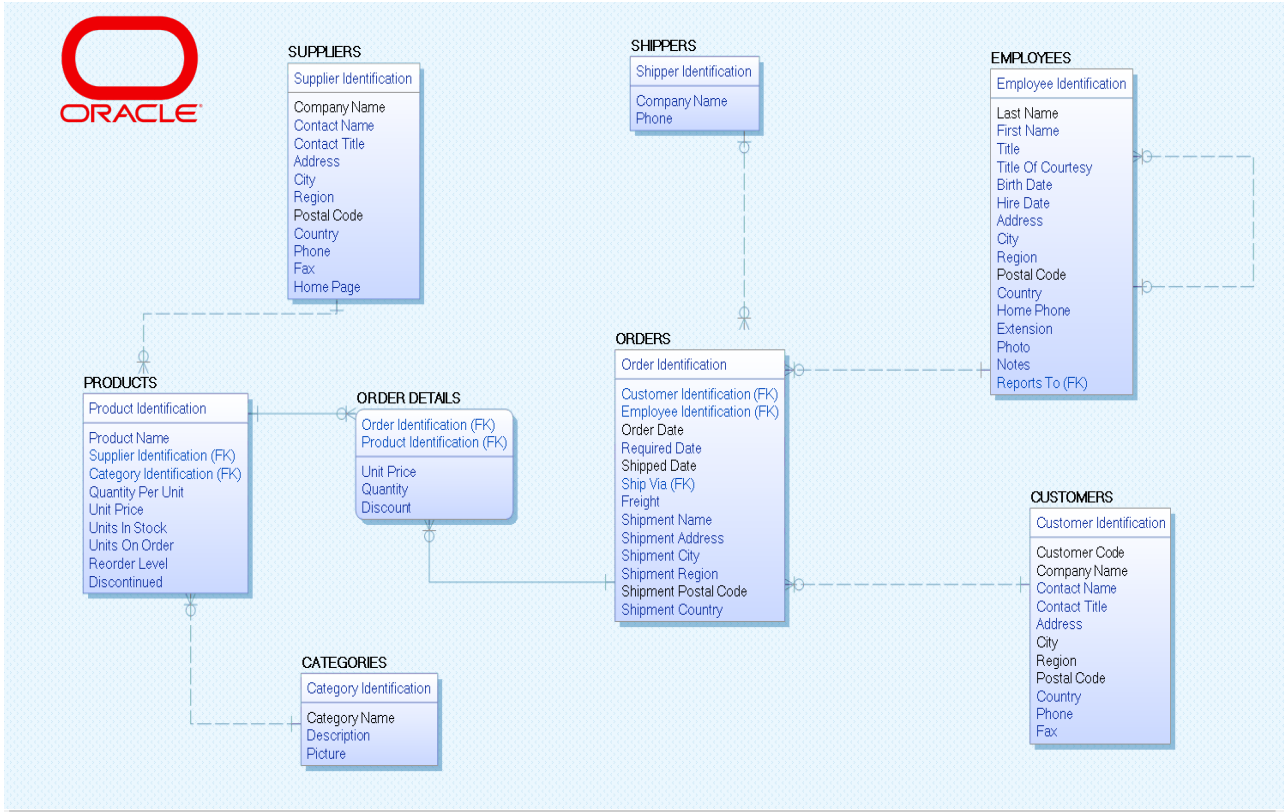
- Convert the Oracle model to a cloud-native database
- Optimize the model for use in a cloud-native database



Forward Engineer

- Generate optimized schema in cloud-native database
- Changes to source Oracle schemas can be easily synced with cloud-native database schemas

Model Before You Move



Data Movement

83%

**of data migration
projects either fail or
exceed their budgets
and schedules.
(Gartner)**

**It's Not Reinvention.
It's Iteration.**

We've been here before.

But now we have better tools — and more expectations.

log-based shipping → streaming CDC
scheduled batch jobs → in warehouse ELT
predefined schema → agile modelling

Same Goals: Quality. Access. Governance.

65%

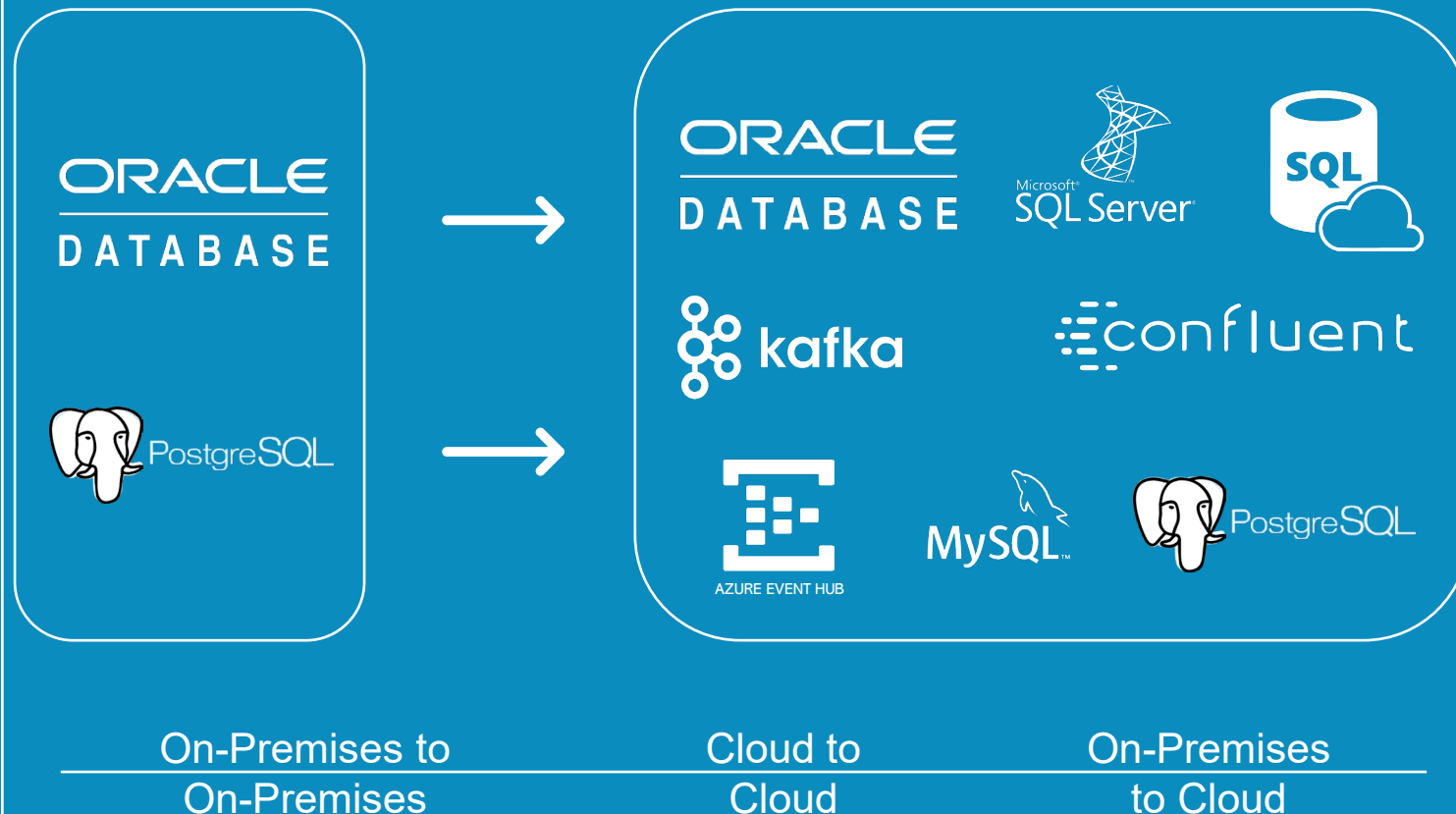
**of data migration
project failures
are due to
schema
mismatches.
(IDC)**

Integration | Replication

Flexible by Nature

“Migrating would have caused about 20 hours of downtime if we'd done it manually. We were able to reduce this time window to just 40 minutes with SharePlex.”

Sunil Nookala
Senior IT Manager, Dell Database Practice



Performance Monitoring & Observability



Benchmarking performance

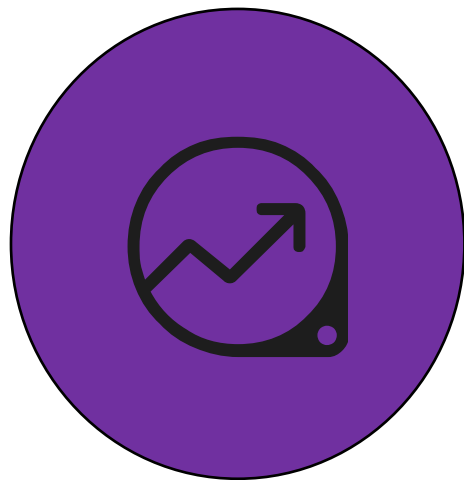
On-premises: Notional cost



Cloud: Real cost



Continuous Monitoring



Poor performance
= increased cost



Avoid throttling



Lower response times

Best practices



Know your
system

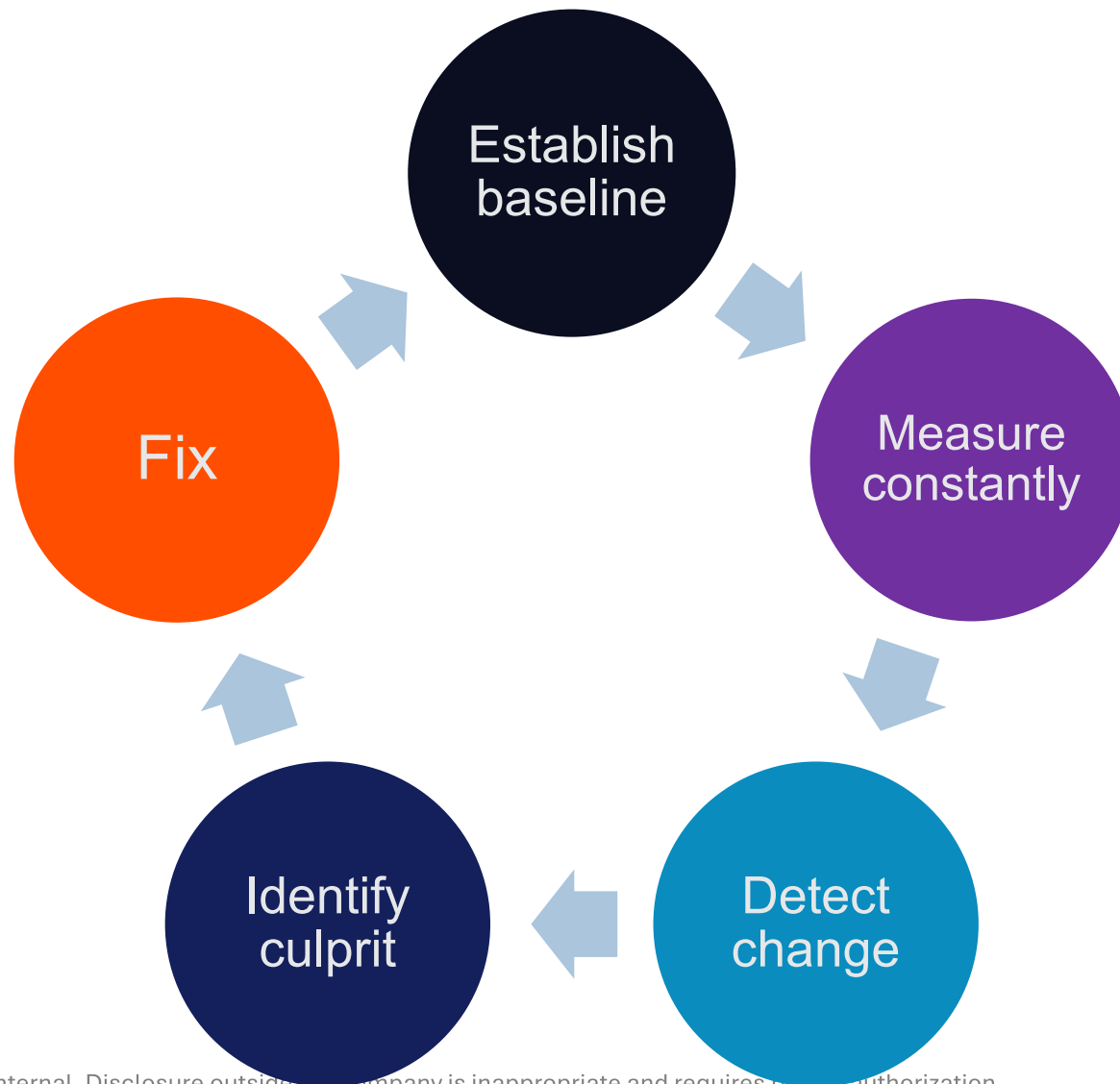


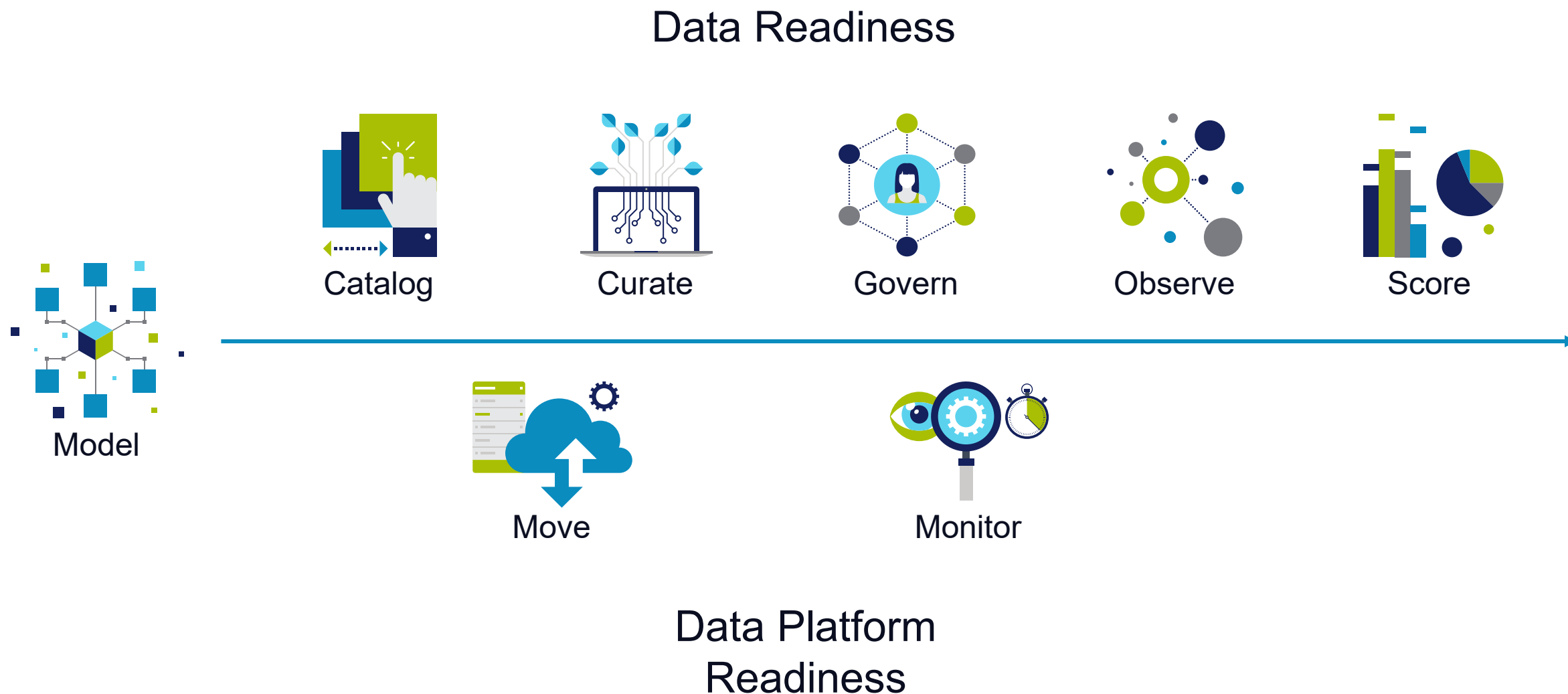
Use the
correct tools



Tune the
workload

Methodically improve performance





Catalog

Build a meta data catalog of all data sources and data assets



Centralized Repository of Metadata

- Scan data sources
- Ingest your Logical & Physical models
- Connect to ETL processes



Understand your Data Inventory

- Understand the **data inventory by Domain**
- Identify **data bloat**
- Review your Logical & Physical models to see **how data is deployed**

Curate

Add context to your metadata



Describe, Define, Tag

- Add descriptions and definitions to your catalog
- Use Tags & Classifications to better categorize the data
- Create & Assign Business Glossary for specific associations



Promote & Increase Data Literacy

- Understand the data in terms of **Business Assets**
- Create narratives necessary for **compliance and your operational processes**



Critical to AI-Readiness

- Gives additional context to AI
- Provides reasoning and rationale to AI-generated actions

Govern

Putting controls and guardrails in place without hindrance



Roles & Responsibilities

- Assign Technical and Business ownership
- Assign Data Stewards, Data Architects, etc.



Associations

- Create Associations between Technical and Business Assets



Contextualized Governance

- The right level of control
- To the right data
- For the right purpose
- At the right time

Observe

End-to-End Data Lineage



Impact Analysis

- Understand the inter-dependencies of your data
- “What will I break if I change this?”
- Vital to understand when tackling modernization and migration efforts



Observe key data pipelines

- Understand where good and bad data is sourced and propagated
- Pinpoint data quality issues where messy re-cursive relationships exist



Include Data Models

- Data models play an important role as they add context to data lineage
- Logical models act as a legend for complex data sources
- The additional metadata inputted into the model can further inform and guide

Score

Comprehensive Data Quality, continuously



Data Profiling & Pattern Analysis

- Monitor Data Accuracy
- Data Completeness
- Data Consistency



Data Drift & Shift

- Detect unexpected data changes
- Detect anomalies
- Prevent downstream disruptions



Data Observability

- Data Volumes
- Data Freshness
- Data Uniqueness

Before
After

Source
Oracle



Target

Snowflake



Quest

Step 0

Impact
Analysis



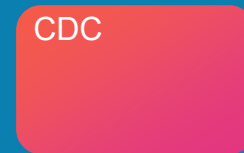
Step 1

Schema
Conversion



Step 2

CDC



Step 3

Workload
Tuning



Data
Governance

Target

Snowflake



Data Integrity



Case Studies

Quest Customers Automating with Quest erwin Platform

Current customers on their journey to trusted data for AI at speed and scale



Insurance (Property & Casualty, Specialty)

Japan's oldest insurance company

51,000 Employees
\$49.7B Revenues



Energy Transmission

One of Europe's largest energy companies

78,000 Employees
\$86B Revenues
(2024)



Electric Utilities

FirstEnergy serves over 6 million customers across six states in the USA

12,000 Employees
\$14B Revenues
(2024)



Banking & Financial Services

217,000 Employees
\$80B Revenues
(2024)

**Customers in production and benefiting from
Quest erwin Data Management Platform - Phase 1**

Building a Trusted Data Ecosystem with erwin Data Platform



About Tokio Marine HCC

Tokio Marine HCC is a leading global specialty insurer with 4,100 employees, A++ ratings, and 100+ products through 60+ acquisitions.

Industry: Financial Services/Insurance

Location: United Kingdom

"When I joined, Tokio Marine had no CDO or existing data office. My first step was listening—spending three months understanding the business and engaging with stakeholders. The executive leadership had already set up a Data Executive Committee, providing a strong head start for governance and alignment..... Today, if it's not in the erwin data marketplace, it doesn't exist."

- Kevin Smith, Chief Data Officer, Tokio Marine HCC

The Challenge & Transformation

Before 2022, Tokio Marine HCC lacked centralized data management and governance. Kevin Smith joined as CDO with a mandate to build a comprehensive system to catalog data and create trusted data products, aiming to establish data as a foundational, accessible resource for decision-making across the complex organization.

Tokio Marine HCC implemented Quest's erwin Data Management Platform to create a self-sufficient, automated data ecosystem. They meticulously designed data products with trust scores, lineage tracking, and built-in governance, ensuring accuracy and consistency. This has significantly boosted confidence in reporting and data utilization across sales, marketing, operations, and other functions.

The Business Impact

Impact from an automated process to build data products

- **750 Days** of combined time saved across multiple business units by replacing manual spreadsheet processes
- **30 Days** of effort delivered transformational business value that previously took months
- **Improved** time to market by building out a process with automation

Thank you



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**Your feedback is
important.**

**Scan this QR Code or
use the Mobile App to
share your thoughts on
this session.**

